Can We See John Stockton's Heart Rate in the Box Score?

3-Post NBA Durability Series — Status 2026-10-09

The Premise

John Stockton and Boris Diaw reportedly share a resting heart rate of ~35 BPM — well below the elite-athlete norm of ~50 BPM. The question: does that physiology leave a detectable trace in box-score data as statistical resilience — fewer missed games, steadier late-game performance, more durable decision-making under fatigue? Stockton's durability is currently circulating on social media, providing a natural hook. ## Post Structure

Post	Subject	Data Window
1	John Stockton + 14 PG comps	E1/E3 full career 1984–2003; E2/E4/E5 1996–2003 only
2	Boris Diaw + 5 PF comps (Nowitzki, Boozer, Jamison, Marion, Lewis)	2004–2016, full BRef
3	All qualifying BRef-era guards + forwards	1980–2010 debuts

Stockton ships first to ride the current conversation; Diaw follows immediately as independent methodological confirmation — different position, different era, full BRef coverage, no age-regime caveat. Post 3 addresses the cherry-picking objection: placing both players in the full distribution of all BRef-era guards and forwards who meet the career-length threshold.

Estimands

	Metric	Data Tier	Status
E1	Games missed per 82 — career durability	Game logs	Code done; fetch partial
E3	Shot selection: 3PA/FGA, FTA/FGA on B2B vs rest	Game logs	Code done; fetch partial
E2	4Q FT% degradation vs Q1–Q3	Quarter box scores	Code done; blocked on E1
E4	4Q AST/TOV ratio vs full-game	Quarter box scores	Code done; blocked on E1
E5	4Q TS% vs Q1–Q3	Quarter box scores	Code done; blocked on E1
E6	Full Q1→Q2→Q3→Q4 scoring profile — shape of fatigue curve	Quarter box scores	Proposed; no new scraping

E1 and E3 are primary. E6 (profile plot) requires no new data — a subplot within E2/E4/E5 analysis.

Inference Approach

E1's denominator corrects for lockout-shortened seasons: 1999→50 games, 2012→66, 2020→65, 2021→72.

n_treated = 1 makes bootstrap CIs invalid. For E1/E3: resample the treatment label across all N cohort players 5,000 times; report where the observed gap falls in the null distribution. For E2/E4/E5: mixed-effects model `metric ~ quarter * is_treated + rest_days + (1|player_id)` via `statsmodels.MixedLM`; permutation resamples `is_treated` at the player level (not game level) to preserve within-player correlation.

Current Status

Infrastructure: complete. 131 tests passing across four scripts:

Script	Purpose
`fetch_bref_data.py`	BRef game log scraper; HTML-cached per (player, season); 4 sec/request
`analyze_durability.py`	E1 + E3; OLS + permutation; emits `{cohort}_comparators.csv`
`fetch_quarter_splits.py`	Quarter box score scraper; cached by game ID; <=3 req/sec
`analyze_quarter_splits.py`	E2/E4/E5; mixed-effects model; player-level permutation

Data fetch: incomplete. The PG fetch ran but stopped short; PF produced zero data.

Status	Players
Full PG data	Stockton, Payton, Jackson, Porter, Thomas, Hardaway, K. Johnson
Partial (1 season only)	Kidd (82 games), Nash (62 games)
Cached but likely below >=28 MPG threshold	Bogues, Strickland
Never fetched — PG	Anderson, Douglas, Harper, Blaylock
PF cohort	0 players — fetch did not run

Caveats (mandatory in every post)

1. **Selection-on-outcome bias** — >=600-game threshold conditions on the dependent variable; E1 estimates durability *among players who completed long careers*, not all players.
2. **B2B state selection** — playing a B2B selects healthier player-states; within-player pairing removes stable effects but not selection.
3. **Blowout/garbage time** — starters sit in non-competitive games; 4Q minutes are not random. Compresses treatment effect toward zero.
4. **Intentional fouling** — close-game FTA inflation in Q4; FT% estimates partly reflect pressure context, not just physical state.
5. **Timeout-adjusted rest** — unmeasured confounder; coaches use Q4 timeouts strategically.
6. **Age-regime** — Stockton's quarter data covers ages 34–40 (career decline). A null within-game result does not contradict a prime-years resilience claim.
7. **Pre-1996 quarter gap** — BRef play-by-play starts 1996-97; only Stockton ages 34–40 are covered. Justin Jacobs (Squared Statistics) has video-reconstructed pre-1996 quarter logs that would extend the window to his prime (ages 22–33). Outreach drafted, not sent.
8. **No causality** — frame consistently as "consistent with" and "does not contradict."

What's Left

Step	Blocker	Est.
Re-run PG + PF game log fetch	None — run now	~3h
E1 + E3 analysis — both cohorts	Game log fetch complete	20 min
Quarter box score scrape — Stockton cohort	E1 must emit `pg_comparators.csv`	~40 min
E2/E4/E5 + E6 profile — Stockton cohort	Quarter scrape complete	30 min
Stockton post — fill results, render	Analysis complete	1 day
Diaw post	Stockton post done	1 day
Population cohort build + data fetch	Independent — long scrape	~half day
Population post	Population analysis complete	1 day
Jacobs outreach (#22) — manual browser form	Human action required	20 min

NOTE: Causal language forbidden. Null results are valid findings — state them plainly. Sourcing asymmetry: Diaw's 35 BPM comes from documented post-retirement interviews; the Stockton figure is less sourced — acknowledge this explicitly in the Stockton post.